Himaayah Schools
ACADEMY

FREE GUIDE

وَأَحْصَىٰ كُلَّ شَيْءٍ عَدَدًا

"And He has counted everything in number." (al-Jinn 72:28)

JAMB Mathematics

A UTME study guide: the syllabus in five areas, every formula you need, worked examples, common traps and a practice test.

بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

Before you begin

Mathematics in the UTME is a test of speed as much as knowledge: forty questions in about twenty-five minutes of your time. Most questions are short calculations that are easy once you know the method. This guide gives you the syllabus in five areas, the formulas you must know by heart, nine worked examples, the traps that cost marks, an 8-week plan and a practice test.

The UTME at a glance	
Subjects	Use of English and three subjects for your course. Mathematics is required for science, engineering, medical, business and social science courses.
Questions	Use of English 60; each other subject 40; 180 questions in all, on computer (CBT).
Time	About two hours for all four papers, so roughly 40 seconds a question. You move freely between subjects.
Marking	Each subject is scored out of 100, for a total out of 400. There is no negative marking: never leave a question blank.

Check the official details

Exam rules and syllabuses can change from year to year. Always check the current JAMB brochure and the official Mathematics syllabus on the JAMB website or e-facility portal, and follow them where they differ from this guide.

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Part 1 The syllabus at a glance

Study from the **JAMB syllabus**, not only from your school notes: every UTME question comes from a topic on it. The topics fall into these areas:

Area	Topics
Number and numeration	Number bases; fractions, decimals and approximations; indices, logarithms and surds; sets; percentages, ratio, simple and compound interest
Algebra	Polynomials and factorisation; linear and quadratic equations; simultaneous equations; variation; inequalities; progressions (AP and GP); binary operations; matrices and determinants
Geometry and trigonometry	Angles and polygons; circles and their theorems; mensuration (areas and volumes); loci; coordinate geometry; trigonometric ratios, sine and cosine rules, bearings, angles of elevation and depression
Calculus	Differentiation of polynomials and simple functions, gradients and rates; maxima and minima; integration of polynomials; area under a curve
Statistics and probability	Mean, median, mode; frequency tables, histograms, pie charts; range, variance and standard deviation; permutations and combinations; probability

Algebra and number questions make up a large share of the paper. Calculus and statistics are very predictable: master them for quick, sure marks.

Part 2 The formula sheet

Number and algebra

Topic	Formula
Indices	$a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$; $a^0 = 1$; $a^{-n} = 1/a^n$; a^(1/n) = $\sqrt[n]{a}$
Logarithms	log(xy) = log x + log y; log(x/y) = log x − log y; log x^n = n log x; $\log_a a = 1$; $\log_a 1 = 0$
Quadratics	$x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$; sum of roots = −b/a; product of roots = c/a
Arithmetic progression	$T_n = a + (n - 1)d$; $S_n = n/2\ [2a + (n - 1)d]$
Geometric progression	$T_n = ar^{n-1}$; $S_n = a(r^n - 1)/(r - 1)$; $S\infty = a/(1 - r)$ for $\lvert r \rvert < 1$
Interest	Simple: I = PRT/100; compound amount: $A = P(1 + R/100)^n$
Variation	Direct y = kx; inverse y = k/x; joint y = kxz; partial y = a + bx

Mensuration and geometry

Shape	Formula
Circle	Area πr^2; circumference $2\pi r$; arc length $(\theta/360) \times 2\pi r$; sector area $(\theta/360) \times \pi r^2$
Cylinder	Volume $\pi r^2 h$; curved surface $2\pi rh$; total surface $2\pi r(r + h)$
Cone	Volume $\frac{1}{3}\pi r^2 h$; curved surface πrl (l = slant height)
Sphere	Volume $\frac{4}{3}\pi r^3$; surface area $4\pi r^2$
Polygons	Sum of interior angles (n − 2) × 180°; each exterior angle of a regular polygon 360°/n
Coordinates	Distance $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$; midpoint $((x_1 + x_2)/2, (y_1 + y_2)/2)$; gradient $(y_2 - y_1)/(x_2 - x_1)$; line y = mx + c
Trigonometry	sin = opp/hyp, cos = adj/hyp, tan = opp/adj; sine rule a/sin A = b/sin B; cosine rule $a^2 = b^2 + c^2 - 2bc \cos A$; area = ½ab sin C

Calculus, statistics and probability

Topic	Formula
Differentiation	$d/dx(x^n) = nx^{n-1}$; d/dx(sin x) = cos x; d/dx(cos x) = −sin x; turning points where dy/dx = 0
Integration	$\int x^n\,dx = x^{n+1}/(n + 1) + c\ (n \neq -1)$; area = ∫ from a to b of y dx
Averages	Mean = Σfx/Σf; median = middle value of ordered data; mode = most frequent
Spread	Variance = $\Sigma f(x - \bar{x})^2/\Sigma f$; standard deviation = √variance
Counting	${}^nP_r = n!/(n - r)!$; ${}^nC_r = n!/[r!(n - r)!]$
Probability	P(E) = favourable/total; P(A or B) = P(A) + P(B) − P(A and B); independent: P(A and B) = P(A) × P(B)

Use π = 22/7 unless the question says otherwise, and keep answers as fractions until the last step: many options are given as fractions.

Part 3 Worked examples

Example 1. Convert 1011_2 to base ten.

→ $1011_2 = 1\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0$

→ $= 8 + 0 + 2 + 1$

Answer: 11

Example 2. Solve $2x^2 - 5x - 3 = 0$.

→ Factorise: $(2x + 1)(x - 3) = 0$

→ $2x + 1 = 0$ or $x - 3 = 0$

Answer: $x = -\frac{1}{2}$ or $x = 3$

Example 3. Evaluate $\log_{10} 8 + \log_{10} 125$.

→ $\log 8 + \log 125 = \log (8 \times 125)$

→ $= \log_{10} 1000 = \log_{10} 10^3$

Answer: 3

Example 4. Find the 5th term of the GP 3, 6, 12, …

→ $a = 3, r = 2$

→ $T_5 = ar^4 = 3 \times 16$

Answer: 48

Example 5. Find the sum of the first 10 terms of the AP 2, 5, 8, …

→ $a = 2, d = 3, n = 10$

→ $S_{10} = 10/2\ [2(2) + 9(3)] = 5 \times 31$

Answer: 155

Example 6. If $y = 3x^3 - 4x + 7$, find dy/dx and the gradient at $x = 1$.

→ $dy/dx = 9x^2 - 4$

→ At $x = 1$: $9 - 4$

Answer: $dy/dx = 9x^2 - 4$; gradient 5

Example 7. Find the area of a sector of radius 7 cm and angle 90°. (π = 22/7)

→ Area $= (90/360) \times (22/7) \times 7^2$

→ $= \frac{1}{4} \times 154$

Answer: 38.5 cm^2

Example 8. Two fair coins are tossed. Find the probability of two heads.

→ Outcomes: HH, HT, TH, TT (4 equally likely)

→ Favourable: HH (1)

Answer: ¼

Example 9. Find the mean and median of 2, 4, 6, 8, 10.

→ Mean = 30 ÷ 5

→ The middle value of the ordered list is the third

Answer: Mean 6, median 6

Part 4 Traps that cost marks

Trap	What to do
Sign errors in algebra	Write each step; check by substituting your answer back into the equation.
Forgetting to order data before finding the median	Always arrange the values first.
Degrees and the calculator mode	Make sure your calculator is in degrees (D) for trigonometry, not radians.
Using the wrong π	Use 22/7 when the radius is a multiple of 7; otherwise use what the question gives.
Confusing permutation and combination	Order matters: permutation. Order does not matter (choosing a team): combination.
Spending three minutes on one question	Skip it, flag it and return. Forty questions in about 25 minutes.
Not using the options	Sometimes substituting the options into the question is the fastest method.

Part 5 An 8-week plan

Start at least eight weeks before the exam. Each week: learn the topics, make short notes and solve past questions under time. Keep one day a week for revision.

Week	Focus	Practice
1	Number bases, indices, logarithms and surds	30 questions a day, timed
2	Equations, factorisation and quadratics	Past questions on each topic
3	Variation, inequalities, progressions	Formula drill, then 30 questions
4	Matrices, sets and binary operations	Past questions; note patterns
5	Mensuration and coordinate geometry	Learn every formula; 30 questions
6	Trigonometry, bearings, circle theorems	Draw diagrams for every question
7	Calculus, statistics and probability	These are sure marks: master them
8	Full revision and mock exams	Two full timed papers; review every error

Exam-day reminders

Read every option before choosing. Eliminate the clearly wrong ones first.

Do not spend more than a minute on one question: flag it and come back.

Answer every question; there is no penalty for a wrong answer.

Check that your answers are saved before you submit, and keep an eye on the clock for all four subjects.

Test Practice test

Give yourself 10 minutes for these 15 questions, as in the real exam. The answers follow.

1. Simplify $2^3 \times 2^4$.

a) 4^7 b) 128 c) 64 d) 2^{12}

2. If 2x + 3 = 11, find x.

a) 7 b) 3 c) 4 d) 5

3. Convert 25 (base ten) to base two.

a) 11001 b) 10101 c) 11011 d) 10011

4. Find the 6th term of the AP 3, 7, 11, ...

a) 19 b) 27 c) 25 d) 23

5. Evaluate $\log_2 32$.

a) 5 b) 16 c) 4 d) 6

6. Find the gradient of the line through (1, 2) and (3, 8).

a) 2 b) 3 c) 6 d) ⅓

7. Find the area of a circle of radius 14 cm. ($\pi = 22/7$)

a) 88 cm² b) 308 cm² c) 616 cm² d) 154 cm²

8. Differentiate x^4.

a) x^3 b) $4x^5$ c) $4x^3$ d) $x^5/5$

9. Find $\int 2x\,dx$.

a) $x^2 + c$ b) $2 + c$ c) $2x^2 + c$ d) $x + c$

10. The probability of a 6 on one throw of a fair die is:

a) ½ b) ⅙ c) ⅓ d) 1

11. Find the mode of 2, 3, 3, 5, 7, 3, 8.

a) 5 b) 7 c) 2 d) 3

12. sin 30° =

a) ½ b) √3/2 c) 1 d) √2/2

13. y varies directly as x, and y = 12 when x = 3. Find y when x = 5.

a) 15 b) 20 c) 18 d) 25

14. In how many ways can the letters of CAT be arranged?

a) 3 b) 9 c) 6 d) 27

15. Find the sum of the interior angles of a hexagon.

a) 540° b) 360° c) 900° d) 720°

No.	Answer	No.	Answer
1	b) 128	9	a) $x^2 + c$
2	c) 4	10	b) ⅙
3	a) 11001	11	d) 3
4	d) 23	12	a) ½
5	a) 5	13	b) 20
6	b) 3	14	c) 6
7	c) 616 cm^2	15	d) 720°
8	c) $4x^3$		

Recommended textbooks

- **New General Mathematics for Senior Secondary Schools**, M. F. Macrae and others.
- Other texts listed in the current JAMB Mathematics syllabus.
- **Past questions**: solve at least the last ten years' UTME Mathematics questions, timing yourself, and study every one you get wrong.

Prepare with the Himaayah Academy

Live UTME, Post-UTME and O-Level classes with a personal mentor, daily tasks, weekly mock exams, CBT practice and a leaderboard, all online.

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وَمَا تَوْفِيقِي إِلَّا بِاللَّهِ

"And my success is only through Allah." (Hūd 11:88)

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Know the method, then go fast.

This free guide from the Himaayah Academy helps UTME candidates prepare for Mathematics with a formula sheet, worked examples and a clear plan.

- The Mathematics syllabus in five areas
- A formula sheet for number, algebra, geometry, calculus and statistics
- Nine worked examples, step by step
- The traps that cost marks
- An 8-week plan and a 15-question practice test

Prepare with the Himaayah Academy

Live UTME classes with a personal mentor, daily tasks, weekly mock exams and CBT practice, all online.

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